

Roll No.

TBT-601

**B. TECH. BIOTECHNOLOGY (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

ANIMAL BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Write the name of Equipments use in Animal cell culture. (CO1)
(b) What is the role of buffer in animal cell culture ? Explain natural buffer. (CO1)
(c) Explain Primary and Secondary cell culture. (CO1)
 2. (a) Give an overview on transgenic mice. (CO2)
(b) What is Embryo transfer technology and how does it help in livestock production ? (CO2)
(c) What is cryopreservation ? (CO1)
 3. (a) Explain embryonic stem cell methods of gene transfer. (CO2)
(b) Give an overview on transgenic pig. (CO2)
(c) What do you understand by oncogenesis ? (CO3)

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(2)

4. (a) What is gene cloning ? Describe with labeled diagram. (CO4)
- (b) What is Transfection ? Explain. (CO3)
- (c) Describe the expression system in mammalian cell culture. (CO4)
5. (a) What is ornamental marine species ? (CO5)
- (b) What is antisense RNA technology ? (CO4)
- (c) What is transgenic fishes ? (CO5)

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B. TECH. BIOTECHNOLOGY (SIXTH SEMESTER) END SEMESTER EXAMINATION, June, 2023

PLANT BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are PGRs ? Explain its type with suitable example. (CO1)
(b) What is clonal propagation ? Write the advantages and limitations of clonal propagation. (CO1)
(c) Explain the setup and requirements of a Plant Tissue Culture laboratory. (CO1)
2. (a) What is somatic embryogenesis ? Write the advantages and limitations of it. (CO1 & CO2)
(b) Explain the method of protoplast isolation and fusion. Write advantages and limitations also. (CO1 & CO2)
(c) What is Artificial seed ? Explain the method of artificial seed production. (CO1 & CO2)

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(2)

3. (a) Discuss the role of Plant Tissue Culture techniques in conservation of threatened plant species. (CO3)
- (b) Write short notes on the following : (CO3)
- (i) Cryopreservation
- (ii) Green revolution in India
- (c) What is Nitrogen fixation ? Explain different methods of Nitrogen fixation. (CO3)
4. (a) What is Genetic transformation ? Explain the methods of genetic transformation. (CO4)
- (b) Explain the role of *Agrobacterium tumefaciens* in genetic transformation. (CO4)
- (c) Write short notes on the following : (CO4)
- (i) T-DNA
- (ii) *Agrobacterium rhizogenes*
5. (a) Application of Plant Tissue Culture techniques in Secondary metabolite production (CO4/CO5)
- (b) Discuss social and legal issues associated with transgenic plants. (CO4/CO5)
- (c) Write short notes on the following : (CO4/CO5)
- (i) Bt-cotton
- (ii) Transgenic plants

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**B. TECH. BIOTECHNOLOGY (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

BIOSEPARATION TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe characteristics of Type I, II and III types of Bioproducts. Give example of each. (CO1)
(b) Explain the production and downstream processing steps in Penicillin production. (CO1)
(c) Discuss the steps involved in the product isolation and purification of an enzyme. (CO1)
2. (a) What are the physical and chemical methods of cell disruption ? (CO2)
(b) Give an outline of the protein precipitation methods. (CO2)
(c) Discuss the principle of constant pressure filtration. Explain continuous rotary vacuum filters. (CO2)
3. (a) What are the various parameters of Chromatography ? (CO3)

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- (b) Explain Electrochromatography. How is it different from membrane chromatography ? (CO3)
- (c) Write a short note on SDS PAGE. Explain composition of SDS PAGE sample buffer. (CO3)
- 4. (a) Classify the membrane separation processes. What are the advantages of the membrane separation processes ? (CO4)
- (b) What is Ultrafiltration ? What factors affect it and how is it different from reverse osmosis ? (CO4)
- (c) Discuss steps of crystallization. What are the different methods of crystallization ? (CO4)
- 5. (a) What are various strategies for waste water treatment adopted in Industry ? (CO5)
- (b) Explain Greener Bioprocessing and Microscale bioprocessing utilized in Bioseparation industry. (CO5)
- (c) How do we treat the effluents from pharmaceutical industry ? (CO5)

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TBT-604(a)

B. TECH. BIOTECHNOLOGY (SIXTH SEMESTER) END SEMESTER EXAMINATION, June, 2023

BIOSIMILAR TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate the ways and application of genomics in biosimilar development. (CO1)
- (b) Explain the regulatory guidelines and its need for biosimilar development. (CO1)
- (c) Describe stepwise the prerequisites of biosimilar and conditions for its approval. (CO1)
2. (a) Write a note on : Vaccine biologics and explain at least five examples of it. (CO2)
- (b) What is stem cell therapy ? Give examples of FDA approved cell-based therapies and their indications. (CO2)

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- (c) "Biosimilars reduces the cost of biologics many fold". Explain using any example. (CO2)
3. (a) Describe the techniques used for adsorption of protein and peptides on different surfaces. (CO3)
- (b) What are the methods of protein sequencing ? Explain them. (CO3)
- (c) What are the factors that impact protein stability and how the stability can be enhanced. (CO3)
4. (a) Discuss the experimental design and statistical issues involved in bioequivalence studies. (CO5)
- (b) What are the requirements for a biosimilar to be designated interchangeable by the FDA ? (CO5)
- (c) Elucidate the role and basis of PURPLE BOOK. (CO5)
5. (a) Discuss a case study for AstraZeneca's long-acting monoclonal antibody combination, AZD7442 (Evusheld) for COVID-19 preexposure prophylaxis in adults and children at least 12 years old. (CO5)
- (b) Highlight a case Study involving the use of streptokinase in acute myocardial infarction. (CO5)
- (c) Discuss a case study for Interferon alpha developed by Zydus Cadila. (CO5)

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END SEMESTER EXAMINATION APRIL 2023

Name of the Program: **B.Tech. Biotechnology**

Semester: **VI**

Name of the Course: **BIOSENSORS AND BIOELECTRONICS**

Course Code: **TBT-604b**

Time: 3 hrs.

Maximum Marks: 100

Note:-

- (i) All the questions are compulsory; each question has three parts (a), (b), and (c), Attempt any two parts of each question
- (ii) Each question carries 20 marks

Q1	(20 marks)	
(a)	Explain the working mechanism of whole cell based biosensor with an example	CO1
(b)	Discuss the applications of biosensors in (i) health care (ii) Agriculture and food (iii) Environment	
(c)	Write down the short notes on following bioreceptors used in biosensors (i) Enzyme (ii) Antibodies (iii) Nucleic acid	
Q2	(20 marks)	
(a)	Discuss the working principle of a Bio-sensor Transducer. What are the characteristics of a transducer in biosensor?	CO1, CO3, CO4
(b)	Explain the various components of biosensor transducer?	
(c)	Describe the working of chemiluminescent-based biosensors?	
Q3	(20 marks)	
(a)	Describe the working principle of SPR based biosensor with an application	CO1/ CO2
(b)	Explain the working principle and application of amperometric biosensor	
(c)	Describe the possible working mechanisms of glucose biosensors	
Q4	(20 marks)	
(a)	Write a short note on design and fabrication of biosensors.	CO1, CO3,
(b)	Explain in detail the criterion of selection of transducer and design of biosensors.	
(c)	Discuss the selection criteria of biosystem and material used in it.	
Q5	(20 marks)	
(a)	What do you understand by bioelectronics? List the various potential advantages of bioelectronics.	CO2, CO4, CO5
(b)	What are molecular wires and switches? Briefly describe the working of these components in bioelectronics.	
(c)	What do you understand by a biomolecular computer? Describe its development and working.	

END SEM Examination - 2023

Name of the Program: B.Tech (Biotech)
Name of the Course: Statistical Data Analysis with R

Semester: VI
Course Code: TCS-471

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1 (20 marks)		CO1
a)	Statistical analysis is required in digital era for giving important insights in business. Justify the statement.	
b)	Find the probability of a 4 turning up at least once in two tosses of a fair die.	
c)	A and B play 12 games of chess, of which 6 are won by A, 4 are won by B, and 2 end in a draw. They agree to play a match consisting of 3 games. Find the probability that (a) A wins all 3 games, (b) 2 games end in a draw, (c) A and B win alternately, and (d) B wins at least 1 game.	
Q2 (20 marks)		CO3
a)	Create user defined function in R for calculating the factorial of any number.	
b)	A man goes fishing. The number of fish he catches in one hour has a Poisson distribution with mean 1.25. He continues fishing for four hours, then goes home, unless he catches five fish before the four hours are up, in which case he goes home as soon as he has caught the fifth fish. Find (a) the probability that he goes home with three fish. (b) the expected number of fish he takes home	
c)	Explain the various steps of Data Exploration & preparation.	
Q3 (20 marks)		CO3
a)	<pre>fruits <- list("apple", "banana", "cherry") for (x in fruits) { if (x == "cherry") { break } print(x) }</pre> What will be the output of this R Program code given above ? Modify this code by replacing break with next, what will be the change in the output.	
b)	Create the numeric matrix in R of elements {2, 4, 6, 8, 7, 9, 5, 4, 2} of 3X3. Show the syntax in R for adding additional rows and column elements one by one {5,6,7} for this existing matrix.	
c)	Training = c("Strength", "Stamina", "Other"), Pulse = c(100, 150, 120), Duration = c(60, 30, 45) You are required to create the data frame of the health data given	